Technological University Dublin - Blanchardstown

Academic term	2022-23
Year of Study	Year 2
Semester	SEMESTER ONE – January 2023
Date of Examination	To be entered by School Admin
Time of Examination	To be entered by School Admin

January 2023

Programme Code	Programme Title	Module Code
TU653	Higher Certificate in Science in Computing in Information Technology	COMP H2026
TU757	Bachelor of Science in Computing in Information Technology	COMP H2026
TU758	Bachelor of Science in Computing in Digital Forensics & Cyber Security	COMP H2026
TU860	Bachelor of Science (Honours) in Computing	COMP H2026
TU863	Bachelor of Science (Honours) in Computing in Digital Forensics & Cyber Security	COMP H2026

Module Title	Information Technology Mathematics

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External Examiner(s): **Dr. Chris Meudec**

Instructions to candidates:

1) **To ensure that you take the correct examination, please check that the module and programme which you are following is listed in the table above.**
2) **Question 1 is Compulsory. Candidates should attempt ALL parts of Question 1 and any other TWO questions.**
3) **This paper is worth 100 marks. Question 1 is worth 40 marks. The remaining questions are worth 30 marks each.**
4) **Remember to mark each number and section clearly and to list the questions which you have attempted in the correct order on the front of your exam script.**
5) **Show all your work.**
6) **This is a 2 hour exam.**

DO NOT TURN OVER THIS PAGE UNTIL YOU ARE TOLD TO DO SO

Question 1: (Compulsory)

Attempt ALL Parts

a) Given the following matrices:

$$\boldsymbol{A} = \begin{bmatrix} \mathbf{3} \\ \mathbf{2} \\ \mathbf{-5} \end{bmatrix} \boldsymbol{B} = \begin{bmatrix} \mathbf{5} & \mathbf{6} & \mathbf{0} \\ \mathbf{-3} & \mathbf{1} & \mathbf{8} \end{bmatrix} \text{ and } \boldsymbol{C} = \begin{bmatrix} \mathbf{1} & \mathbf{-2} & \mathbf{0} \\ \mathbf{5} & \mathbf{1} & \mathbf{2} \end{bmatrix}$$

i. What is the *size* of each matrix *A* and *B*?
ii. Write down the value of the following elements: b_{22} and a_{11}.
iii. Find the matrix given by *B - 3C*

(5 marks)

b) Evaluate the *determinant* of the matrix *A* below:

$$A = \begin{bmatrix} 2 & -1 & 1 \\ 4 & -2 & 3 \\ 4 & -1 & 5 \end{bmatrix}$$

(5 marks)

c) Evaluate *t*(2) and *t*(3) and *t*(4) for the following recursively defined sequence:

$$t(1) = 2$$
$$t(n) = t(n-1) + 3 \qquad (n>1)$$

(5 marks)

d) Draw the graph *G* represented by the following adjacency matrix and write down the degree of each vertex in *G*.

$$\begin{vmatrix} 0 & 1 & 0 & 1 & 0 \\ 1 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 & 1 \\ 1 & 1 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 & 0 \end{vmatrix}$$

(5 marks)

e) Using the data 12, 10, 18, 14, 8, 10 compute the

i. Mean
ii. Standard Deviation

(5 marks)

f) If two fair 6-sided dice are thrown:

i. What is the probability of getting a 1 and a 4?
ii. What is the probability that the SUM of the numbers will be greater than or equal to 3? **(5 marks)**

g) Using differentiation from **first principles**, find the derivative of $\boldsymbol{y = 4x^2 - 2x + 3}$ **(5 marks)**

h) Evaluate the following *indefinite* integrals:

i. $\int \left(\boldsymbol{4x^3 + 3x^2 - 2x + 7}\right). \boldsymbol{dx}$
ii. $\int \left(\boldsymbol{x^{-3}}\right). \boldsymbol{dx}$ **(5 marks)**

Total: 40 marks

Question 2: Differentiation & Integration

a) If two variables x and y are related by the equation $y = 3x^2 - 2x$, calculate the **average rate of change** of y with respect to x as x varies from 1 to 2. **(5 marks)**

b) Given the function: $\boldsymbol{y = 4x^3 - 12x + 2}$.

i. Find $\frac{dy}{dx}$ and $\frac{d^2y}{dx^2}$. **(5 marks)**

ii. Find the slope of the tangent line to the curve at the point where $\boldsymbol{x = 0}$. **(2 marks)**

iii. Find the coordinates of the *maximum* and *minimum* turning points of the curve. **(6 marks)**

c) A circuit is designed so that it outputs an alternating current (*I*) whose value in milliamps at any time *t* seconds after the circuit is activated is given by: $1 + 20t - 5t^2$. Find,

i. The current 1 second after the circuit is activated.

ii. The rate at which the current is changing after 1.5 seconds.

iii. The time taken for the current to reach its first peak value. **(6 marks)**

d) Find the area enclosed by the curve $y = 2x^2 + 3x + 1$ for x between 1 and 3. **(6 marks)**

Total: 30 marks

Question 3: Statistics & Probability

a) The number of computers in eight rooms are:
3, 15, 1, 2, 1, 30, 2, 1

i. What is the *median* number of computers? **(2 marks)**
ii. What is the *modal* number of computers? **(1 mark)**
iii. What would be the most sensible measurement to use to get the average of this dataset – *mean*, *mode* or *median*? Give reason(s) for your answer. **(2 marks)**
iv. Construct a box plot for this dataset. **(4 marks)**

b) The following is the distribution of the number of books bought by 20 people in the past year.

Table 3.1

Number of books	Frequency (No. of people)
0.0 – 4.0	2
5.0 – 9.0	4
10.0 – 14.0	6
15.0 – 19.0	8

For the grouped data presented in Table 3.1:

i. Calculate the **Mean** of the grouped data. **(3 marks)**
ii. Calculate the **Standard Deviation** of the grouped data. **(5 marks)**
iii. Draw a **Histogram** of the grouped data. **(4 marks)**
iv. Comment on the **symmetry** or **skewness** of the distribution of the histogram in part iii. **(2 marks)**

c) In a café, an average of 3 out of 5 customers ask for water with their food. In a random sample of 10 customers, what is the probability, using the *binomial distribution*, that exactly 6 ask for water with their food? **(7 marks)**

Total: 30 marks

Question 4: Matrices, Graphs & Trees

a) Let $\boldsymbol{A} = \begin{bmatrix} \mathbf{4} & \mathbf{2} \\ \mathbf{3} & \mathbf{1} \end{bmatrix}$ and $\boldsymbol{B} = \begin{bmatrix} \mathbf{9} & \mathbf{10} \\ \mathbf{2} & \mathbf{5} \end{bmatrix}$

i. Write down the *Identity Matrix* of *B* **(2 marks)**

ii. Evaluate $(-2B^T + AB)$ **(6 marks)**

iii. Calculate the *inverse* of the matrix *B* **(5 marks)**

b) Describe the difference between *breadth-first* and *depth-first* graph traversals. Consider the following graph *G = (V, E)* (figure 4.1), starting at node *c*, in what order will the nodes be visited using breadth-first search? Assume, that if there is a choice between nodes, the algorithm will choose alphabetically.

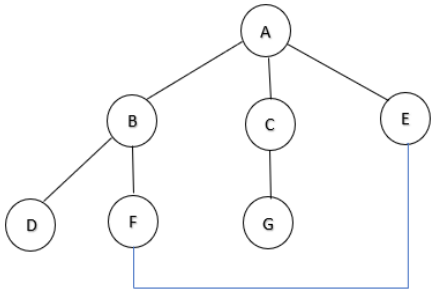


Figure 4.1 Graph G

(9 marks)

c) For the weighted graph in Figure 4.2 below, use *Prim's algorithm*, beginning with vertex *A,* to find a *Minimal Spanning Tree* and give its *weight*.

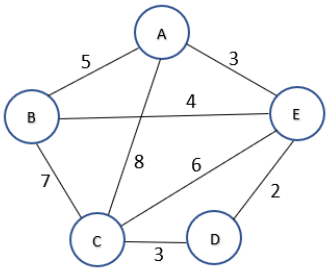


Figure 4.2

(8 marks)

Total:30 Marks

Formulae

Matrix Determinant

$\det A = ad - bc$ (2x2 matrix)

$\det A = a_{11}c_{11} + a_{12}c_{12} + a_{13}c_{13}$ (3x3 matrix using row 1)

Matrix Inverse

$A^{-1} = \frac{1}{\det A}\begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$ (2x2 matrix)

$A^{-1} = \frac{1}{\det A}C^T$, where *C* is the matrix of Cofactors (3x3 matrix)

General Probability:

If two events A and B are mutually exclusive, then the following law holds:

$P[A \text{ or } B] = P[A] + P[B]$

If the two events are not mutually exclusive, the law above is broadened so that the probability of A or B occurring is given by:

$P[A \text{ or } B] = P[A] + P[B] - P[A \text{ and } B]$

Consider two events *A* and *B* with event *B* dependent on *A*. The probability of the occurrence of both events is given by:

$P[A \text{ and } B] = P[A] \times P[B \mid A]$

Probability Distributions:

$P[X = r] = {}^nC_r\, p^r q^{n-r} = \frac{n!}{(n-r)!r!} p^r q^{n-r}$ (Binomial distribution)

$P[X = r] = \frac{e^{-\mu}\mu^r}{r!}$, where $\mu = np$ (Poisson distribution)

$a = \frac{x - \mu}{\sigma}$ (Normal distribution)

Statistics:

$$\bar{x} = \frac{\sum_{i=1}^{n} x_i}{n}$$ (Dataset mean)

$$s = \sqrt{\frac{\sum_{i=1}^{n} (x_i - \bar{x})^2}{n-1}}$$ or $$s^2 = \frac{\sum_{i=1}^{n} x_i^{\,2} - n\left(\bar{x}\right)^2}{n-1}$$ (Dataset standard deviation)

$$\bar{x} = \frac{\sum_{i=1}^{n} f_i m_i}{\sum_{i=1}^{n} f_i}$$ (Grouped data mean)

$$s = \sqrt{\frac{\sum_{i=1}^{n} f_i (m_i - \bar{x})^2}{\sum_{i=1}^{n} f_i - 1}}$$ or $$s^2 = \frac{\sum_{i=1}^{M} f_i m_i^{\,2} - M\left(\bar{x}\right)^2}{M-1}$$ (Grouped standard deviation)

Derivation:

First Principles: $$\frac{dy}{dx} = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h}$$ **Power Rule:** $$\frac{dy}{dx} = nx^{n-1}$$

Product Rule: $$\frac{df}{dx} = u\frac{dv}{dx} + v\frac{du}{dx}$$ **Quotient Rule:** $$\frac{df}{dx} = \frac{v\frac{du}{dx} - u\frac{dv}{dx}}{v^2}$$

Integral: $$\int ax^n dx = \frac{ax^{n+1}}{n+1} + c, \quad n \neq -1$$

Definite Integral:

$$\int_a^b f'(x)dx = f(x)\Big]_a^b = f(b) - f(a)$$

Table of Probabilities – P[*Z* > a], where *Z* is the standard normal variable.

Z	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
0.0	0.5000	0.4960	0.4920	0.4880	0.4840	0.4801	0.4761	0.4721	0.4681	0.4641
0.1	0.4602	0.4562	0.4522	0.4483	0.4443	0.4404	0.4364	0.4325	0.4286	0.4247
0.2	0.4207	0.4168	0.4129	0.4090	0.4052	0.4013	0.3974	0.3936	0.3897	0.3859
0.3	0.3821	0.3783	0.3745	0.3707	0.3669	0.3632	0.3594	0.3557	0.3520	0.3483
0.4	0.3446	0.3409	0.3372	0.3336	0.3300	0.3264	0.3228	0.3192	0.3156	0.3121
0.5	0.3085	0.3050	0.3015	0.2981	0.2946	0.2912	0.2877	0.2843	0.2810	0.2776
0.6	0.2743	0.2709	0.2676	0.2643	0.2611	0.2578	0.2546	0.2514	0.2483	0.2451
0.7	0.2420	0.2389	0.2358	0.2327	0.2296	0.2266	0.2236	0.2206	0.2177	0.2148
0.8	0.2119	0.2090	0.2061	0.2033	0.2005	0.1977	0.1949	0.1922	0.1894	0.1867
0.9	0.1841	0.1814	0.1788	0.1762	0.1736	0.1711	0.1685	0.1660	0.1635	0.1611
1.0	0.1587	0.1562	0.1539	0.1515	0.1492	0.1469	0.1446	0.1423	0.1401	0.1379
1.1	0.1357	0.1335	0.1314	0.1292	0.1271	0.1251	0.1230	0.1210	0.1190	0.1170
1.2	0.1151	0.1131	0.1112	0.1093	0.1075	0.1056	0.1038	0.1020	0.1003	0.0985
1.3	0.0968	0.0951	0.0934	0.0918	0.0901	0.0885	0.0869	0.0853	0.0838	0.0823
1.4	0.0808	0.0793	0.0778	0.0764	0.0749	0.0735	0.0721	0.0708	0.0694	0.0681
1.5	0.0668	0.0655	0.0643	0.0630	0.0618	0.0606	0.0594	0.0582	0.0571	0.0559
1.6	0.0548	0.0537	0.0526	0.0516	0.0505	0.0495	0.0485	0.0475	0.0465	0.0455
1.7	0.0446	0.0436	0.0427	0.0418	0.0409	0.0401	0.0392	0.0384	0.0375	0.0367
1.8	0.0359	0.0351	0.0344	0.0336	0.0329	0.0322	0.0314	0.0307	0.0301	0.0294
1.9	0.0287	0.0281	0.0274	0.0268	0.0262	0.0256	0.0250	0.0244	0.0239	0.0233
2.0	0.0228	0.0222	0.0217	0.0212	0.0207	0.0202	0.0197	0.0192	0.0188	0.0183
2.1	0.0179	0.0174	0.0170	0.0166	0.0162	0.0158	0.0154	0.0150	0.0146	0.0143
2.2	0.0139	0.0136	0.0132	0.0129	0.0125	0.0122	0.0119	0.0116	0.0113	0.0110
2.3	0.0107	0.0104	0.0102	0.0099	0.0096	0.0094	0.0091	0.0089	0.0087	0.0084
2.4	0.0082	0.0080	0.0078	0.0075	0.0073	0.0071	0.0069	0.0068	0.0066	0.0064
2.5	0.0062	0.0060	0.0059	0.0057	0.0055	0.0054	0.0052	0.0051	0.0049	0.0048
2.6	0.0047	0.0045	0.0044	0.0043	0.0041	0.0040	0.0039	0.0038	0.0037	0.0036
2.7	0.0035	0.0034	0.0033	0.0032	0.0031	0.0030	0.0029	0.0028	0.0027	0.0026
2.8	0.0026	0.0025	0.0024	0.0023	0.0023	0.0022	0.0021	0.0021	0.0020	0.0019
2.9	0.0019	0.0018	0.0018	0.0017	0.0016	0.0016	0.0015	0.0015	0.0014	0.0014
3.0	0.0013	0.0013	0.0013	0.0012	0.0012	0.0011	0.0011	0.0011	0.0010	0.0010
3.1	0.0010	0.0009	0.0009	0.0009	0.0008	0.0008	0.0008	0.0008	0.0007	0.0007
3.2	0.0007	0.0007	0.0006	0.0006	0.0006	0.0006	0.0006	0.0005	0.0005	0.0005
3.3	0.0005	0.0005	0.0005	0.0004	0.0004	0.0004	0.0004	0.0004	0.0004	0.0003
3.4	0.0003	0.0003	0.0003	0.0003	0.0003	0.0003	0.0003	0.0003	0.0003	0.0002
3.5	0.0002	0.0002	0.0002	0.0002	0.0002	0.0002	0.0002	0.0002	0.0002	0.0002
3.6	0.0002	0.0002	0.0001	0.0001	0.0001	0.0001	0.0001	0.0001	0.0001	0.0001
3.7	0.0001	0.0001	0.0001	0.0001	0.0001	0.0001	0.0001	0.0001	0.0001	0.0001
3.8	0.0001	0.0001	0.0001	0.0001	0.0001	0.0001	0.0001	0.0001	0.0001	0.0001
3.9	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
4.0	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000